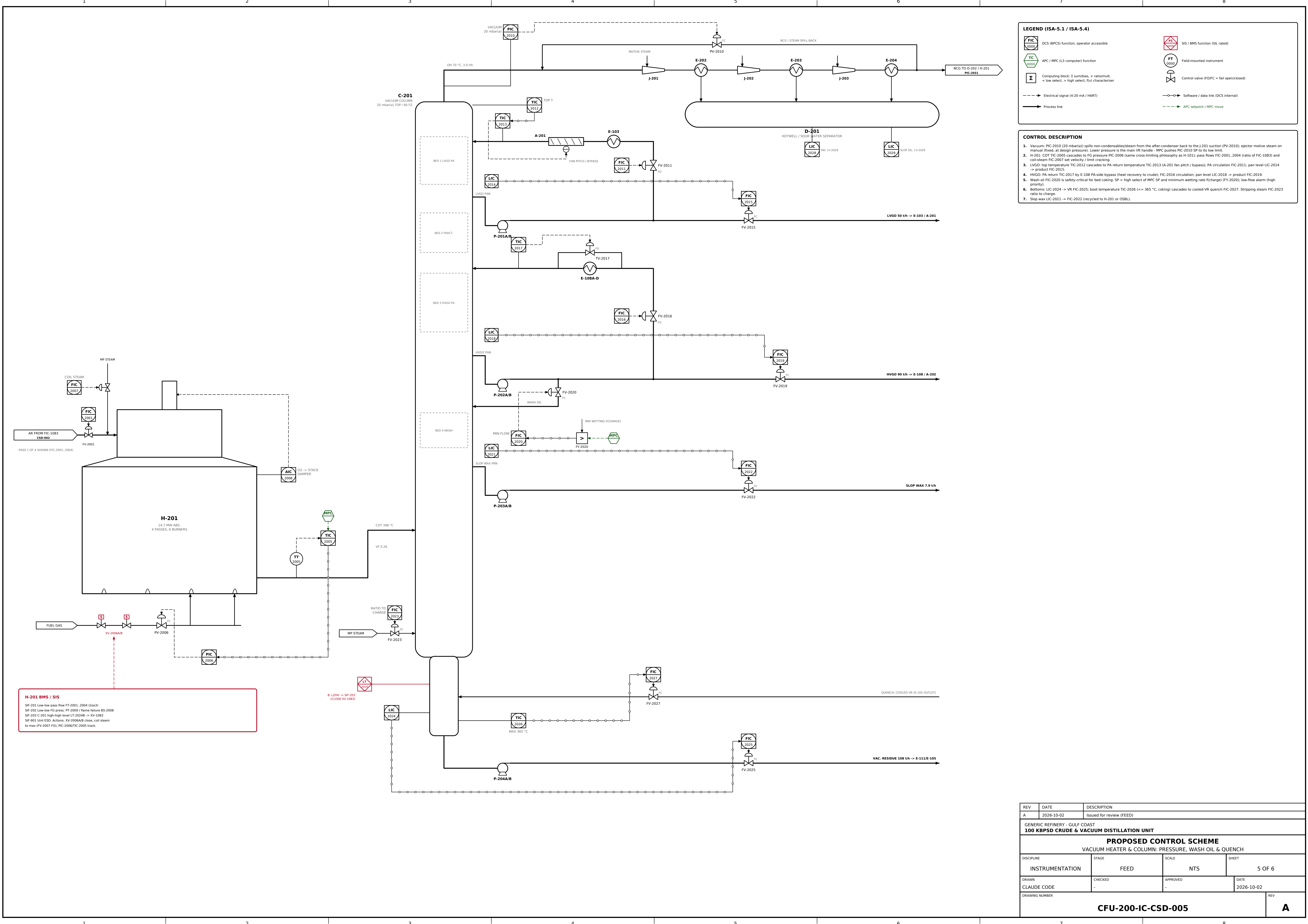




LEGEND (ISA-5.1 / ISA-5.4)

FIC

2000

DCS (BPCS) function, operator accessible

TC

0000

APC / MPC (L3 computer) function

Σ

Computing block: Σ sum/bias, $\times$ ratio/mult, $<$ low select, $>$ high select, (fix) characteriser

Electrical signal (4-20 mA / HART)

Process line

SIS / BMS function (SIL rated)

Field-mounted instrument

Control valve (FO/FC = fail open/closed)

Software / data link (DCS internal)

APC setpoint / MPC move

- CONTROL DESCRIPTION
1. Vacuum: PIC-2010 (20 mbar(a)) spills non-condensables/steam from the after-condenser back to the J-201 suction (PV-2010); ejector motive steam on manual (fixed, at design pressure). Lower pressure is the main lift handle - MPC pushes PIC-2010 SP to its low limit.

2. H-201: COT TIC-2005 cascades to FG pressure PIC-2006 (same cross-limiting philosophy as H-101); pass flows FIC-2001..2004 (ratio of FIC-1083) and coil-steam FIC-2007 set velocity / limit cracking.

3. LVGO: top temperature TIC-2012 cascades to PA return temperature TIC-2013 (A-201 fan pitch / bypass); PA circulation FIC-2011; pan level LIC-2014 -> product FIC-2015.

4. HVGO: PA return TIC-2017 by E-108 PA-side bypass (heat recovery to crude); FIC-2016 circulation; pan level LIC-2018 -> product FIC-2019.

5. Wash oil FIC-2020 is safety-critical for bed coking: SP = high select of MPC SP and minimum wetting rate f(charge) (FY-2020); low-flow alarm (high priority).

6. Bottoms: LIC-2024 -> VR FIC-2025; boot temperature TIC-2026 ($\leq$ 365 °C, coking) cascades to cooled-VR quench FIC-2027. Stripping steam FIC-2023 ratio to charge.

7. Slop wax LIC-2021 -> FIC-2022 (recycled to H-201 or OSBL).

H-201 BMS / SIS

SIF-201 Low-low pass flow FT-2001..2004 (2x3)

SIF-202 Low-low FG press. PF-2009 / flame failure BS-2008

SIF-203 C-201 high-high level LT-2024B -> XV-1083

SIF-901 Unit ESD. Actions: XV-2006A/B close, coil steam to max (PV-2007 FO), PIC-2006/TIC-2005 track.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROPOSED CONTROL SCHEME			
VACUUM HEATER & COLUMN: PRESSURE, WASH OIL & QUENCH			
DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	5 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-200-IC-CSD-005			A